DARBY HUYE

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EDUCATION

PhD Tufts University, Computer Science — Jan 2021 - May 2026
Advisor: Raja Sambasivan

MS Tufts University, Computer Science — Jan 2021
Advisor: Raja Sambasivan

BS Tufts University, Computer Science & Mathematics — May 2020
Advisors: Ming Chow & Misha Kilmer

RESEARCH INTERESTS

Distributed Systems, Observability , Visualization, Performance Debugging

RESEARCH EXPERIENCE

Meta, New York City, NY — 2022-2023
Research Intern, Mentor: Yuri Shkuro

- Characterized Meta's microservice architecture, analyzing the scale, complexity, and topology of its services and how they evolve over time
- Examined request workflow processing, quantifying variability in workflow structures and system behavior
- Published findings at ATC '23, revealing previously undisclosed aspects of Meta's microservices and tracing challenges

Tufts University, Medford, MA
Advisor: Raja Sambasivan

- **Harnessing the power of holes in distributed traces** — 2024-2026
    - Developed low-overhead methods (e.g. bloomfilters) to accurately preserve topological and ordering information across data loss in traces.
    - Built a reconstructor which extracts compact breadcrumbs from collected trace data and uses it to reconstruct accurate topologies and critical paths
- **Correcting errors in Alibaba's open-sourced distributed traces** — 2023
    - Identified two key categories of inconsistencies in Alibaba's trace data, invalidating prior assumptions in research

- Designed an algorithm to remedy errors, ensuring trace reconstruction prioritizes accuracy
- Open-sourced corrected traces and evaluated the impact of different error-handling techniques used in prior research

- **Understanding microservice architectures** 2021-2022
  - Conducted interviews with industry practitioners to analyze real-world perceptions and challenges in microservices
  - Evaluated academic microservice testbeds, identifying key design decisions and their implications
  - Highlighted mismatches between industry-scale microservices and the limited design space of academic testbeds
- **Experiments with academic microservice testbeds** 2020
  - Investigated the complexity DeathStarBench's architecture using distributed tracing
  - Found that traces collected via the provided HTTP workload generators were mostly homogeneous and did not capture the complete and expected functionality of the application

TEACHING & MENTORSHIP

MIT Primes Mentor 2021 – 2023

- Advise high school students in computer science research

Students Advised

Adrita Samanta & Henry Han, "Visualizing Distributed Traces in Aggregate," 2023
Anshul Rastogi & Joey Dong, "Locating Regions of Uncertainty in Distributed Systems using Aggregate Trace Data," 2022
Anshul Rastogi & Tanmay Gupta, "Threshold-Based Inference of Dependencies in Distributed Systems," 2021

PUBLICATIONS

Conference Papers

Zabcic-Matic, T, Huye, D, Lan, L., Sambasivan, R., "Bridges as a primitive for accommodating data loss in distributed tracing," Conditionally accepted at ACM Symposium on Cloud Computing (SoCC'26). DOI: coming soon

Astolfi, T., Silai, V., Huye, D, Lan, L., Sambasivan, R., Bater, J., "Dynamic read & write optimization with TurtleKV," accepted at 2026 VLDB DOI: coming soon

Huye, D, Lan, L., Sambasivan, R., "Systemizing and mitigating topological inconsistencies in Alibaba's microservice call-graph datasets," 2024 ACM/SPEC ICPE 2024. DOI: https://dl.acm.org/doi/10.1145/3629526.3645043

Huye, D., Shkuro, Y., Sambasivan, R., "Lifting the veil on Meta's microservice architecture: Analyses of topology and request workflows," 2023 USENIX Annual Technical Conference (USENIX ATC 23). DOI: https://www.usenix.org/system/files/atc23-huye.pdf

Toslali, M., Ates, E., Ellis, A., Zhang, Z., Huye, D Liu, L., Puterman, S., Coskun, A., Sambasivan, R., "Automating instrumentation choices for performance problems in distributed applications with VAIF," Proceedings of the 12th ACM Symposium on Cloud Computing (SoCC'21). November 1st to November 3rd, 2021. DOI: https://doi.org/10.1145/3472883.3487000

Journal Publications

Huye, D.*, Seshagiri, V.*, Lan, L., Wildani, A., and Sambasivan, R., "Identifying mismatches between microservices testbeds and industrial perceptions of microservices," Journal of Systems Research, vol. 2, no. 1, 2022. DOI: https://doi.org/10.5070/SR32157839 *Contributed Equally

Toslali, M., Ates, E., Huye, D., Zhang, Z., Liu, L., Puterman, S., Coskun, A., Sambasivan, R., "VAIF: Variance-based Automated Instrumentation Framework," Operating Systems Review, vol. 56, no. 1, 2022, pp. 42-50. DOI: https://doi.org/10.1145/3544497.3544504

TALKS

Research Presentation, "The power of holes: Adding holes as a first class primitive in distributed tracing" to industry folks from Meta and Grafana. December 2024.

Guest Lecture, "Introduction to distributed traces" in Cloud Computing at Tufts University. November 2024.

Conference Presentation, "Systemizing and mitigating topological inconsistencies in Alibaba's microservice call-graph datasets," ACM/SPEC ICPE 24. May 2024.

Research Presentation, "Towards correcting incomplete observability data" to industry folks from Meta and Grafana. December 2023.

Conference Presentation, "Lifting the veil on Meta's microservice architecture: Analyses of topology and request workflows," USENIX ATC 23. July 2023.

Guest Lecture, “Intelligent trace sampling strategies” in Debugging Cloud Computing at Tufts University. March 2023.

Research Presentation, “Identifying mismatches between microservice testbeds and industrial perceptions of microservices,” Distributed Systems & Networks Group at Tufts University. April 2022.

Guest Lecture, “Performance Debugging on Microservices with Distributed Tracing” in Debugging Cloud Computing at Tufts University. December 2021.

Research Presentation, “LeitMotif: a tool for discovering emergent communication patterns in microservice applications,” to industry folks from RedHat, Meta, and Grafana. November 2021.

PROFESSIONAL AFFILIATIONS

ACM, 2021-Present

PC Member: ACM/SPEC ICPE 2024 Data Challenge Track